

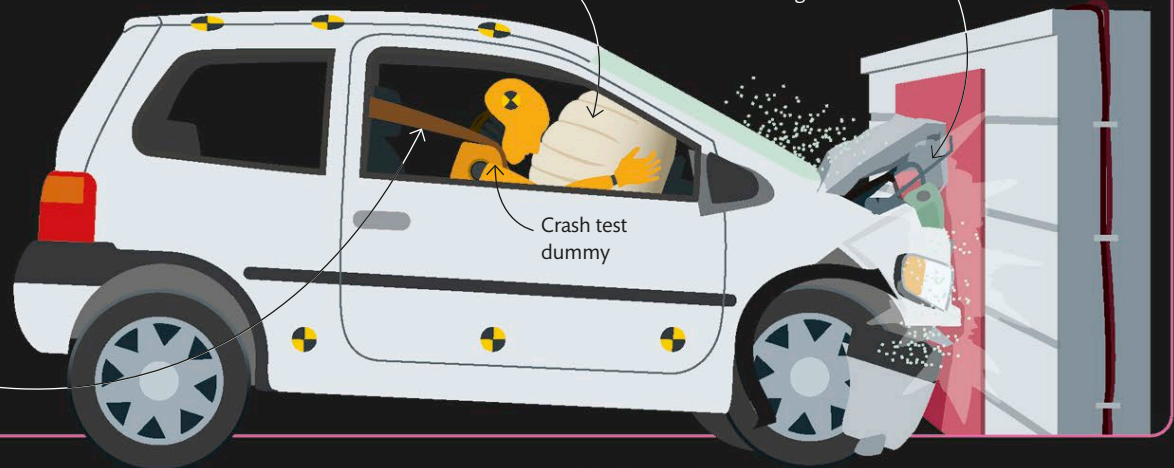
MOMENTUM

Momentum is a measure of how much movement an object has. The faster an object moves, and the more mass it has, the greater its momentum. When a car crashes, its momentum changes quickly, but the passengers keep moving, which puts them in danger. Airbags and seatbelts safely change passengers' momentum, and crumple zones slow the change in the car's momentum.

Airbags safely reduce the momentum of the dummy's head, helping reduce injury.

A crumple zone at each end of the car absorbs much of the impact during a collision, slowing the car's change in momentum.

A seatbelt stretches to slow down the movement of the dummy on impact.



Simulating accidents

Vehicles collide at a motor racing track near Paris, France. This test was carried out without using seatbelts in the backseat and was designed to show why seatbelts are essential and encourage car manufacturers to improve the safety of their design.